

Session 100%

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📊 Your Score: 100

Max Score: 200

Q2. Sum of All Subarrays

Solved

 **Stuck somewhere?**
Ask for help from a TA and get it resolved.

Get help from TA.

Problem Description

You are given an integer array **A** of length **N**.
You have to find the sum of all subarray sums of **A**.
More formally, a subarray is defined as a contiguous part of an array which we can obtain by deleting zero or more elements from either end of the array.
A subarray sum denotes the sum of all the elements of that subarray.

Problem Constraints

1 <= N <= 10⁵
1 <= A_i <= 10⁴

Input Format

The first argument is the integer array **A**.

Output Format

Return a single integer denoting the sum of all subarray sums of the given array.

Example Input

Input 1:
A = [1, 2, 3]

Input 2:
A = [2, 1, 3]

Example Output

Output 1:
20

Output 2:
19

Example Explanation

Explanation 1:
The different subarrays for the given array are: [1], [2], [3], [1, 2], [2, 3], [1, 2, 3].
Their sums are: 1 + 2 + 3 + 3 + 5 + 6 = 20

Explanation 2:
Similiar to the first example, the sum of all subarray sums for this array is 19.

See Expected Output

Java 8 (Array Support)

1 /*
2 * Time : O(N)
3 */
4 import java.util.Scanner;
5
6 public class Solution
7 {
8
9 public static void main(String[] args)
10 {
11 Scanner sc = new Scanner(System.in);
12 String line1 = sc.nextLine();
13 String[] str1 = line1.split(" ");
14 int[] input = new int[str1.length];
15
16 for(int i = 0; i < str1.length; i++)
17 {
18 input[i] = Integer.parseInt(str1[i]);
19 }
20
21 System.out.print(subarraySum(input));
22 }
23 }
24

Test Output Wrong Answer!

Test Test With Custom Input Submit